The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2020-2021

MATHEMATICS FOR COMPUTER SCIENTISTS (COMP1017)

Time allowed TWO hours only

Candidates must NOT start writing their answers until told to do so

There are FOUR questions in the exam paper. Answer all FOUR questions.

This is an OPEN-BOOK exam. Students may use available resources including Internet, textbooks, and lecture slides to help construct their solutions.

Students are not allowed to communicate with each other through whatsoever means during the examination. Any such communication will be deemed as collusion and may be subject to Academic Misconduct discipline according to Quality Manual.

DO NOT turn examination paper over until instructed to do so

1. Questions on Sets and Functions. (20 marks)

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

(i) $\{0,2,12,36,80\} = \{y \in \mathbb{N} \mid \exists x \in \mathbb{N}((y = x^2 + x^3) \wedge (0 \leq x \leq 5))\}$.

(ii) Floor is a function from integers to reals.

(iii) $|\{\{1,2,5\},\{1,2,5\},9,\{\}\}| = 4$.

(iv) $\emptyset \subseteq \emptyset$.

(v) $|2^S| \geq |S|$.

(b) Consider the following design of an abstract memory addressing system. You have a multidimensional array of five dimensions. Assume that along each dimension you have 10 elements. The addressing on a specific location is thus $(a_1, a_2, a_3, a_4, a_5)$. The address for the first location is $(0,0,0,0,0)$ while that for the last location is $(n,n,n,n,n), n > 0$. First, construct the complete addresses as a generalized Cartesian product of sets. Second, provide a brief argument why each address generated will be distinct. Third, calculate the size of this array (i.e. how many addresses in total) from your construction. Also show how much of an increase in the array size when each dimension is increased equally from n to $45n$. Full marks for complete construction using basic concepts and definitions in set theory.

[10 marks]

(c) In your basic construction above, the addresses are elements of a set, which by basic definition has no ordering. Briefly discuss how these addresses can be positioned, which requires imposing some ordering. Higher marks for more formal construction using basic concepts. Hint: consider the notion of a Cartesian coordinate system to represent the position, e.g., $\boldsymbol{x} = (x_1, x_2)$ and Manhattan distance $\mathrm{d}(\boldsymbol{x}, \boldsymbol{y}) = |x_1 - y_1| + |x_2 - y_2|$.

[5 marks]

2. Questions on Induction and Proof. (30 marks)

(a) Consider this cake cutting and coloring problem. The cake is a perfect circular object. Every new cut is made so that it cuts through the center of the cake and at different angular position. For example, if one cuts a cake horizontally at 0 degrees, then no future cuts are made at the same position. Prove using an induction approach that the cake with finite number of cuts can always be colored black and white alternately. First, specify how the number of cuts is represented and what is your induction hypothesis. Second, construct your proof by induction. For this proof approach, full marks are given for answers further supported by way of graphical illustration.

[18 marks]

(b) For the problem above, instead of using graphical illustration, conjecture a formula that can measure the size of the problem. Then use a proof by induction to verify your conjecture for the induction hypothesis. Full marks for complete proof.

[8 marks]

(c) Briefly discuss what differentiates the two proof approaches used.

[4 marks]

3. Questions on Counting and Probabilities. (25 marks)

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

(i) There are 125 different ways of lining up 3 people out of 5.

(ii) Choosing 8 students out of 100 students to form a committee is a combination problem.

(iii) If there are 30 students in a class, then at least two have last names that begin with the same letter.

(iv) Throw an unbiased coin 5 times and the probability of obtaining 3 heads is $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$.

(v) Put 5 apples into 2 boxes. The probability of one of the boxes having exactly 2 apples is 1.

(b) There are 400 students in a programming class. On the midterm test each student received one of the following grades: $A, B, C,$ or D.

(i) There are less students born in the first half of a year than in the latter. 60% of the grade A students were born in the first half of a year. Show that the event that a randomly chosen student was born in the first half of a year is not independent of the event that a randomly chosen student obtained grade A.

[4 marks]

(ii) It turned out that the number of students receiving the grade A, B, C, and D are 95, 120, 110, 75, respectively. What is the probability of 10 randomly chosen students having 3 A's, 3 B's, 2 C's, and 2 D's? Explain.

[6 marks]

(c) A Machine Learning algorithm was developed to predict whether an input image contains a human face or not. It was tested on 800 independently randomly selected images and the prediction errors were recorded. An error occurs when an input image does not contain a human face and the algorithm predicts otherwise, or when an input image contains a human face and the algorithm predicts otherwise.

(i) Identify all the relevant random variables and their respective sample spaces in the scenario given above.

[2 marks]

(ii) Express the prediction error as a random event and provide a formula to calculate its probability based on the frequentist's definition of probability.

[3 marks]

(iii) Out of the 800 testing images, 30% contain a human face for which 20% were predicted wrongly. On the other hand, 65% of the images that contain no human faces were predicted correctly. Now the algorithm predicts for a new unseen image that it contains a human face. What is the probability that the new unseen image indeed contains a human face?

[5 marks]

4. Questions on Numbers and Relations. (25 marks)

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

(i) $\lceil \lfloor x \rfloor \rceil = \lfloor x \rfloor$ for all real numbers x.

(ii) -67 is congruent to 3 modulo 7.

(iii) $gcd(100, 104) = 2$.

(iv) If $a|bc$, where $a, b,$ and c are positive integers and $a \neq 0$, then $a|b$ or $a|c$.

(v) For integers $a, b, m \neq 0,\ a \equiv b \ (mod\ m)$ is a reflexive relation between a and b.

(b) Let $\emptyset$ be an empty set. Show that the relation $R = \emptyset$ on a nonempty set S is symmetric and transitive, but not reflexive.

[6 marks]

(c) Let R be the relation $\{(1,2), (1,3), (2,3), (2,4), (3,1)\}$, and let S be the relation $\{(2,1), (3,1), (3,2), (4,2)\}$. Find $S \circ R$.

[4 marks]

(d) How many nonzero entries does the matrix representing the relation R on $A = \{1, 2, 3, \ldots, 1000\}$ consisting of the first 1000 positive integers have if R is

(i) $\{(a, b) \mid a \leq b\}$?

(ii) $\{(a, b) \mid a = b \pm 1\}$?

(iii) $\{(a, b) \mid a + b = 1000\}$?

(iv) $\{(a, b) \mid a + b \leq 1001\}$?

(v) $\{(a, b) \mid a \neq 0\}$?

Sketch the matrix for each of the above relations.

[10 marks]